Calibration results
==========
Normalized Residuals

Reprojection error (cam0): mean 0.31433673629828757, median 0.27254565466914715, std: 0.20116728158056676
Gyroscope error (imu0): mean 0.10023960152773685, median 0.06098666651170393, std: 0.27579360160966077
Accelerometer error (imu0): mean 0.09959908702612985, median 0.05900951075532079, std: 0.21247193365300912

Residuals

Reprojection error (cam0) [px]: mean 0.31433673629828757, median 0.27254565466914715, std: 0.20116728158056676
Gyroscope error (imu0) [rad/s]: mean 0.0015574266239365235, median 0.0009475522316814436, std: 0.004285016014747183
Accelerometer error (imu0) [m/s^2]: mean 0.031223034708922622, median 0.018498723808447713, std: 0.06660722258808831

Transformation (cam0):

T_ci: (imu0 to cam0):
[[0.99902539 -0.04349734 0.00750055 0.00897419]
[0.01398352 0.15071648 -0.98847813 -0.0039629]
[0.04186572 0.98761963 0.15117783 -0.03244298]
[0. 0. 0. 1.]]

T_ic: (cam0 to imu0):
[[0.99902539 0.01398352 0.04186572 -0.00755178]
[-0.04349734 0.15071648 0.98761963 0.03302895]
[0.00750055 -0.98847813 0.15117783 0.00092011]
[0. 0. 0. 1.]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
0.008742745544266923

Gravity vector in target coords: [m/s^2]
[-0.11374056 -9.80272635 -0.24908269]

Calibration configuration
==========

cam0

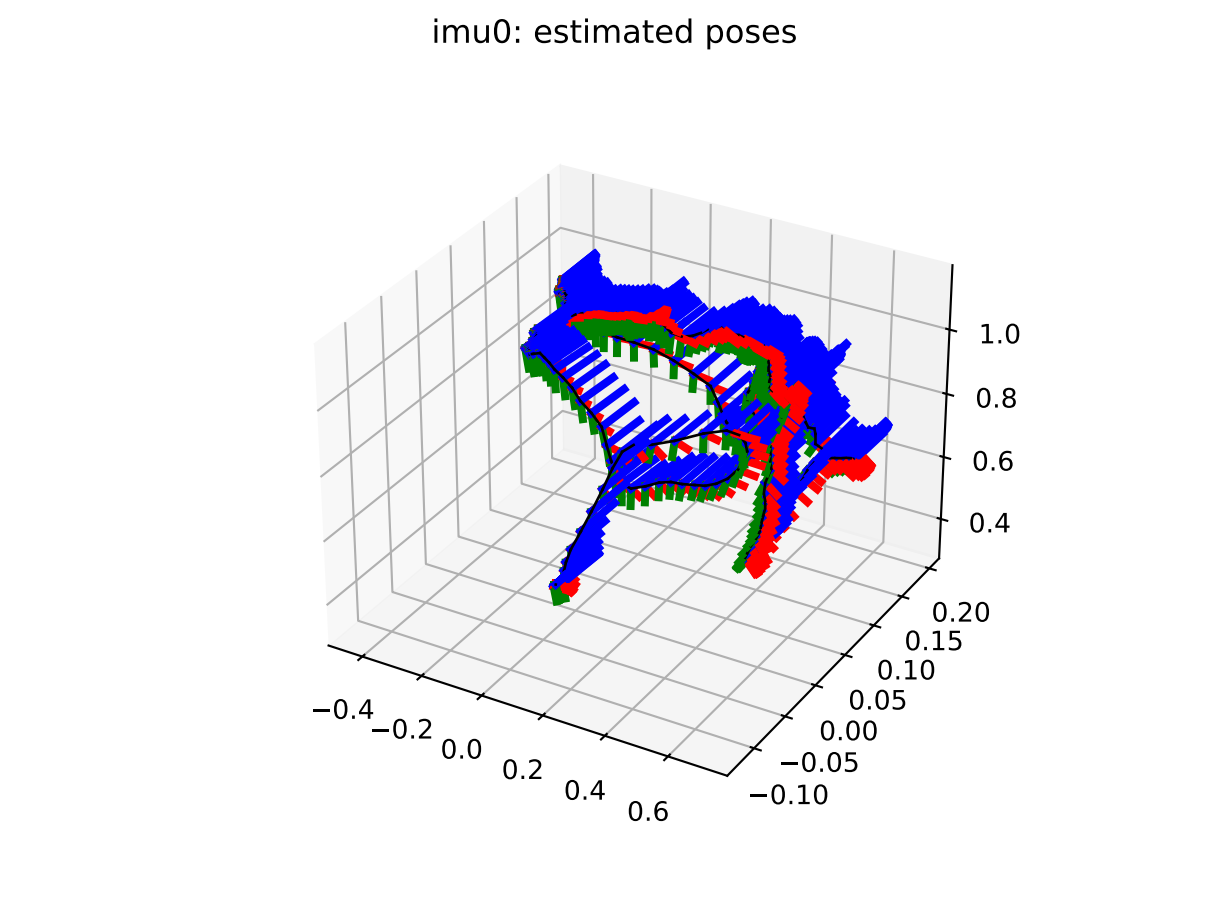
  Camera model: pinhole
  Focal length: [652.879962242561, 652.6913800309958]
  Principal point: [402.7459765146756, 293.51689967360625]
  Distortion model: radtan
  Distortion coefficients: [0.19004630188120233, -0.3566296810567926, -0.001643546078163314,
0.0007208195618292788]
  Type: aprilgrid
  Tags:
    Rows: 6
    Cols: 6
    Size: 0.02 [m]
    Spacing 0.006 [m]

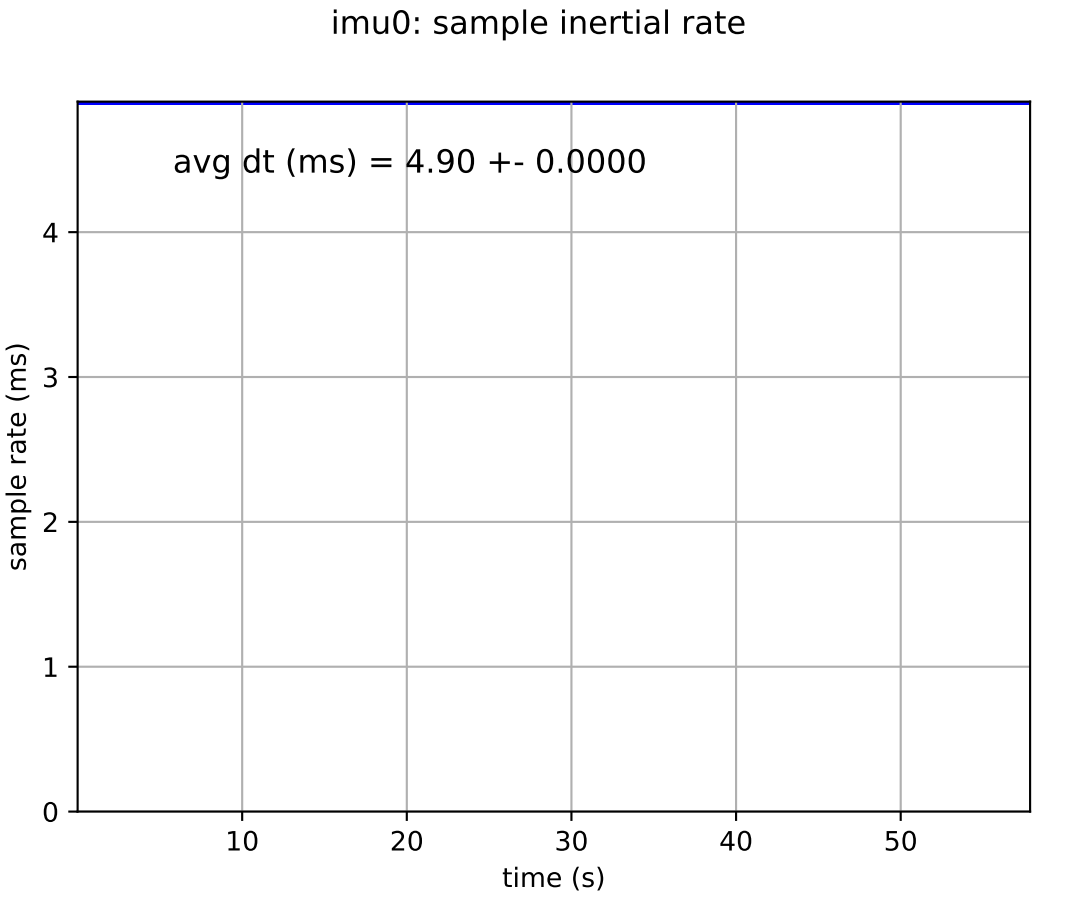
IMU configuration
==========

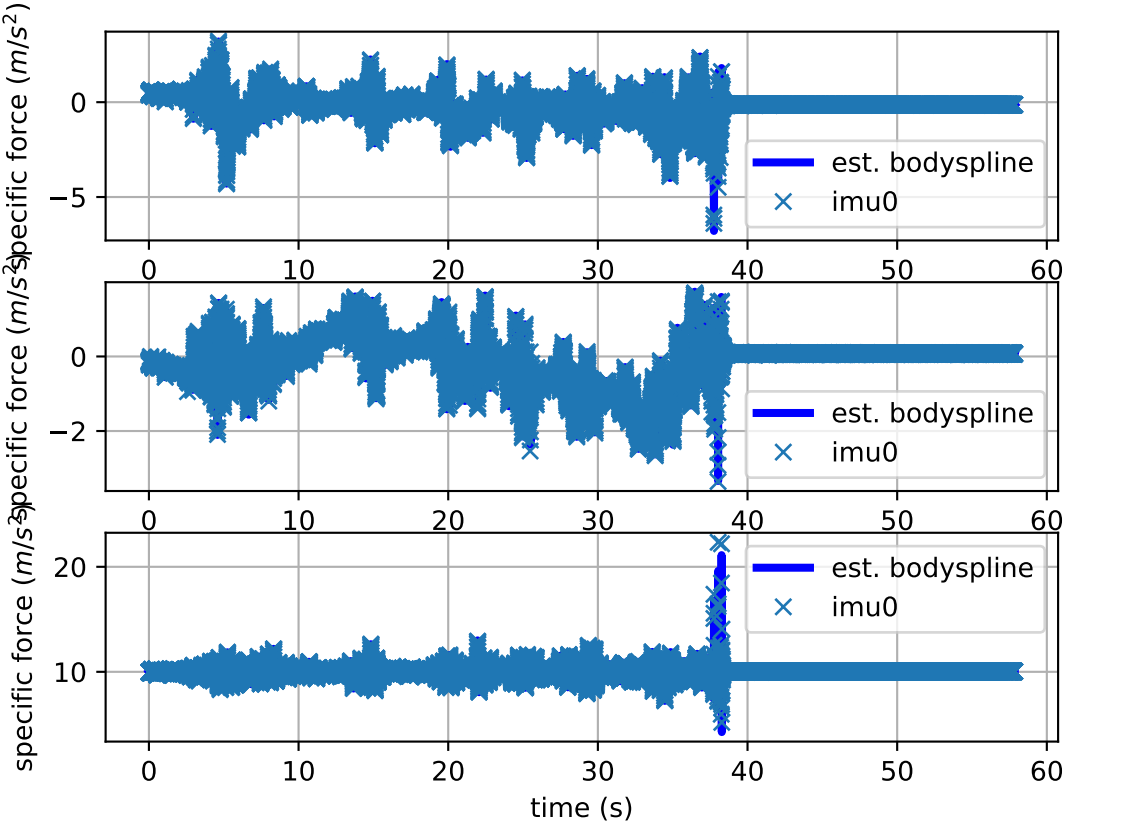
IMU0:

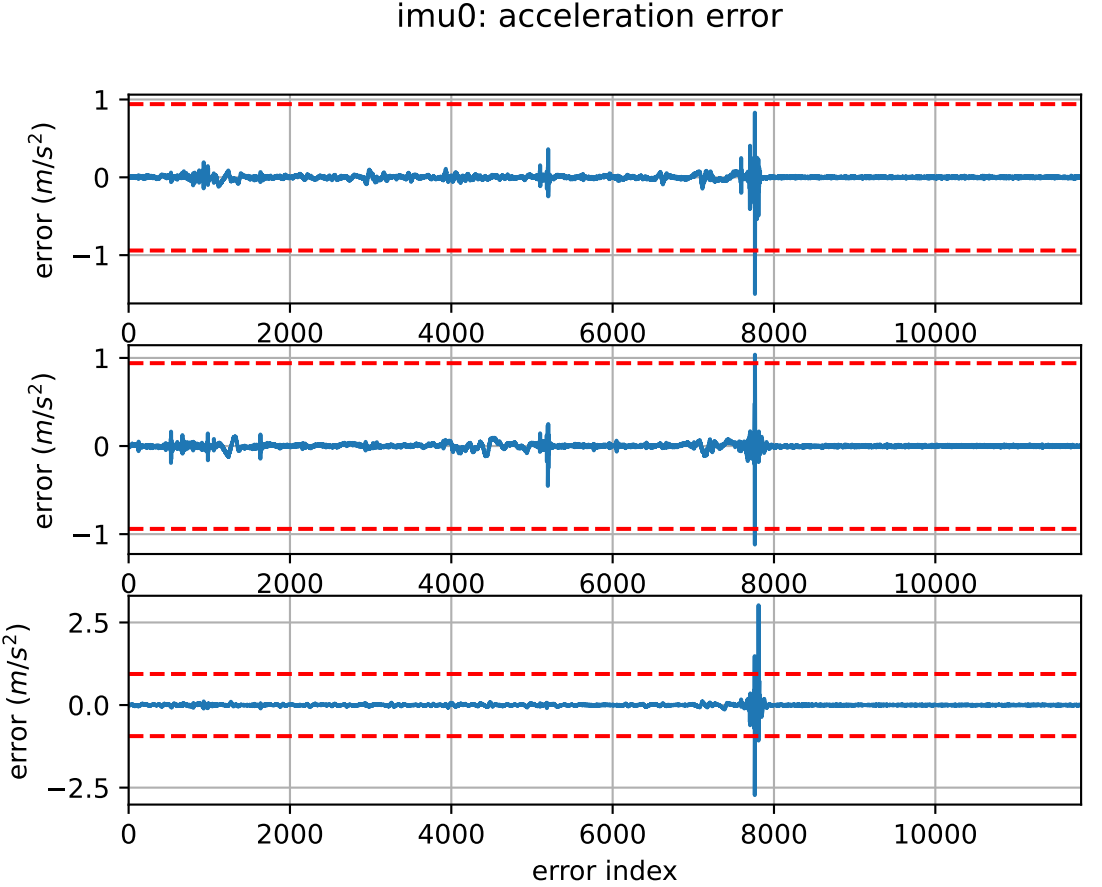
  Model: calibrated
  Update rate: 204.0355927763645
  Accelerometer:
    Noise density: 0.021946576707293677
    Noise density (discrete): 0.3134871577761677
    Random walk: 0.013079819165366822
  Gyroscope:
    Noise density: 0.0010877154466348196
    Noise density (discrete): 0.01553703925594292
    Random walk: 0.00011172744905164805

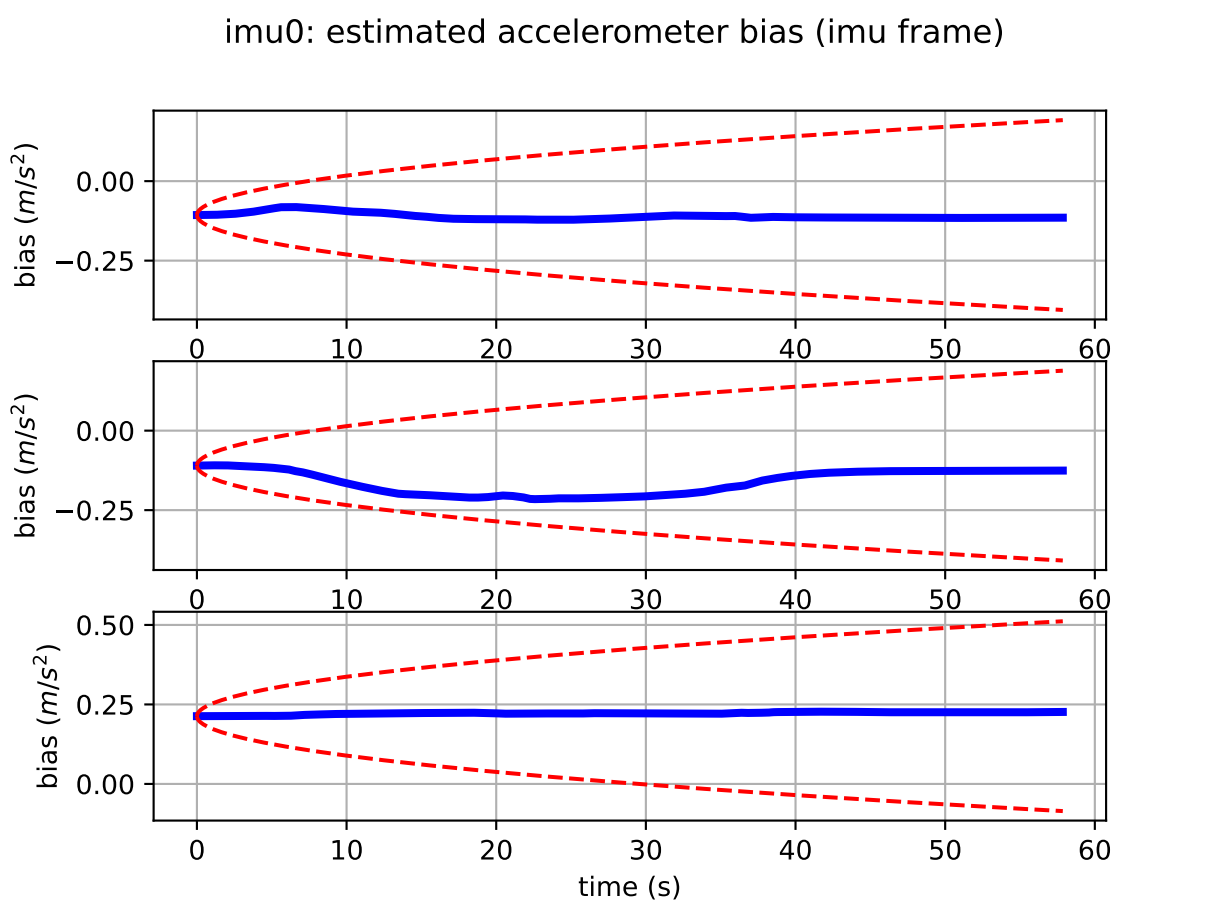
T_ib (imu0 to imu0)
  [[1. 0. 0. 0.]
   [0. 1. 0. 0.]
   [0. 0. 1. 0.]
   [0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]

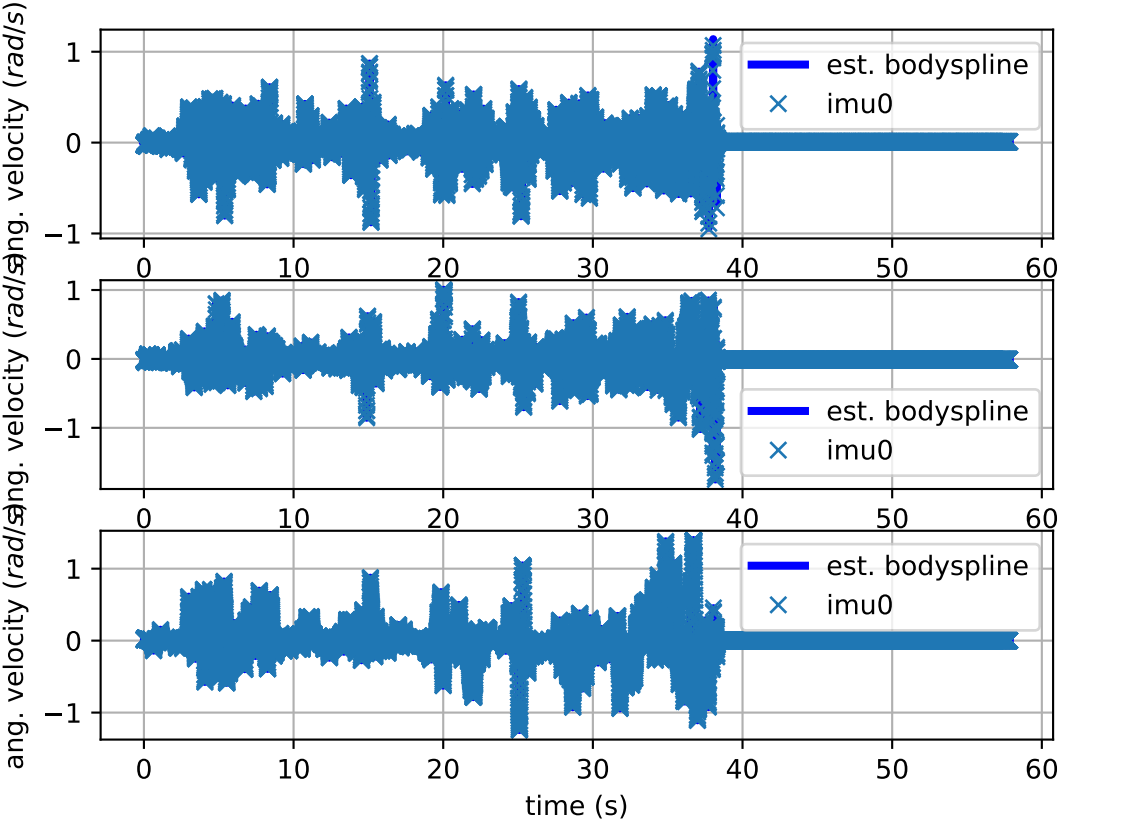

imu0: estimated poses
1.0
0.8
0.6
0.4
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.4
−0.2
0.0
0.2
0.4
0.6

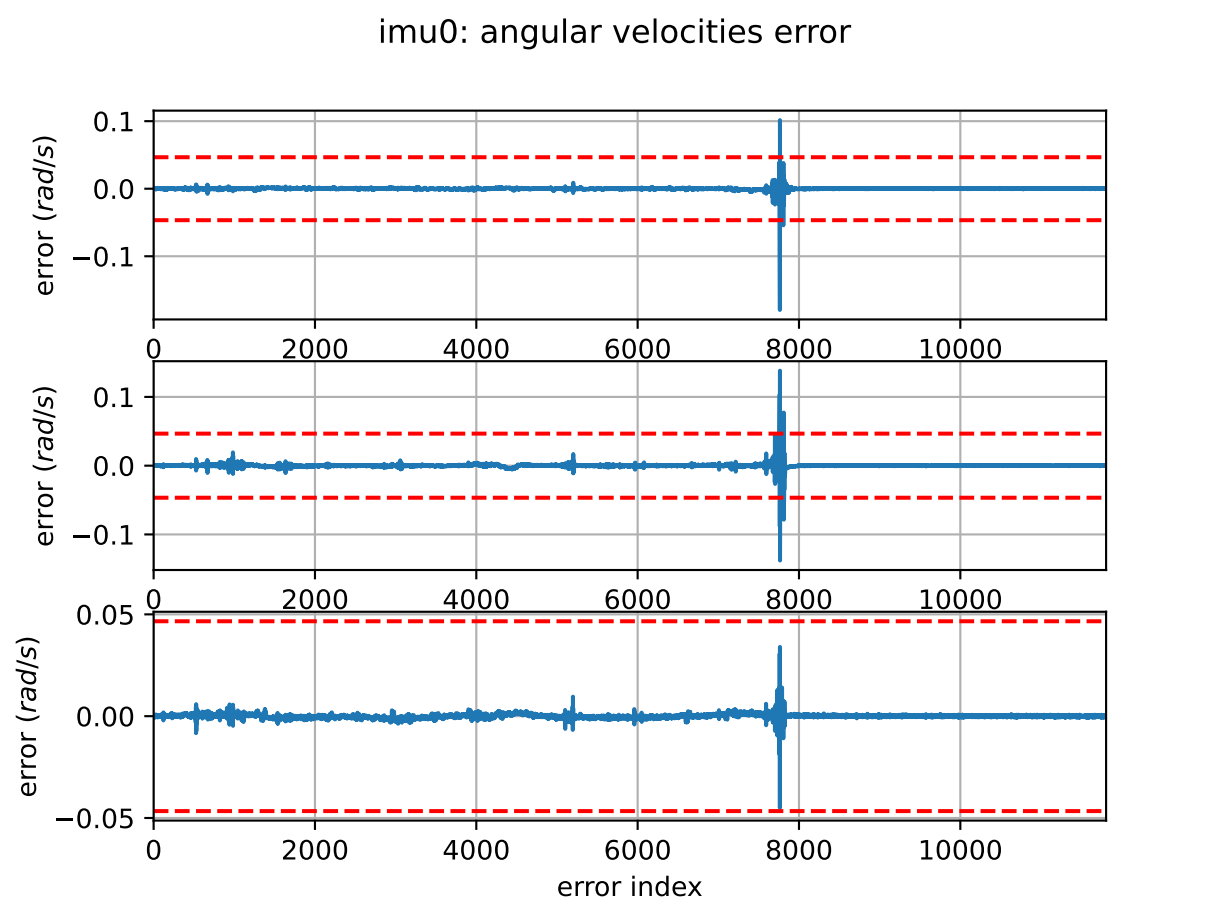

imu0: sample inertial rate
avg dt (ms) = 4.90 +- 0.0000
sample rate (ms)
0
1
2
3
4
10
20
30
40
50
time (s)

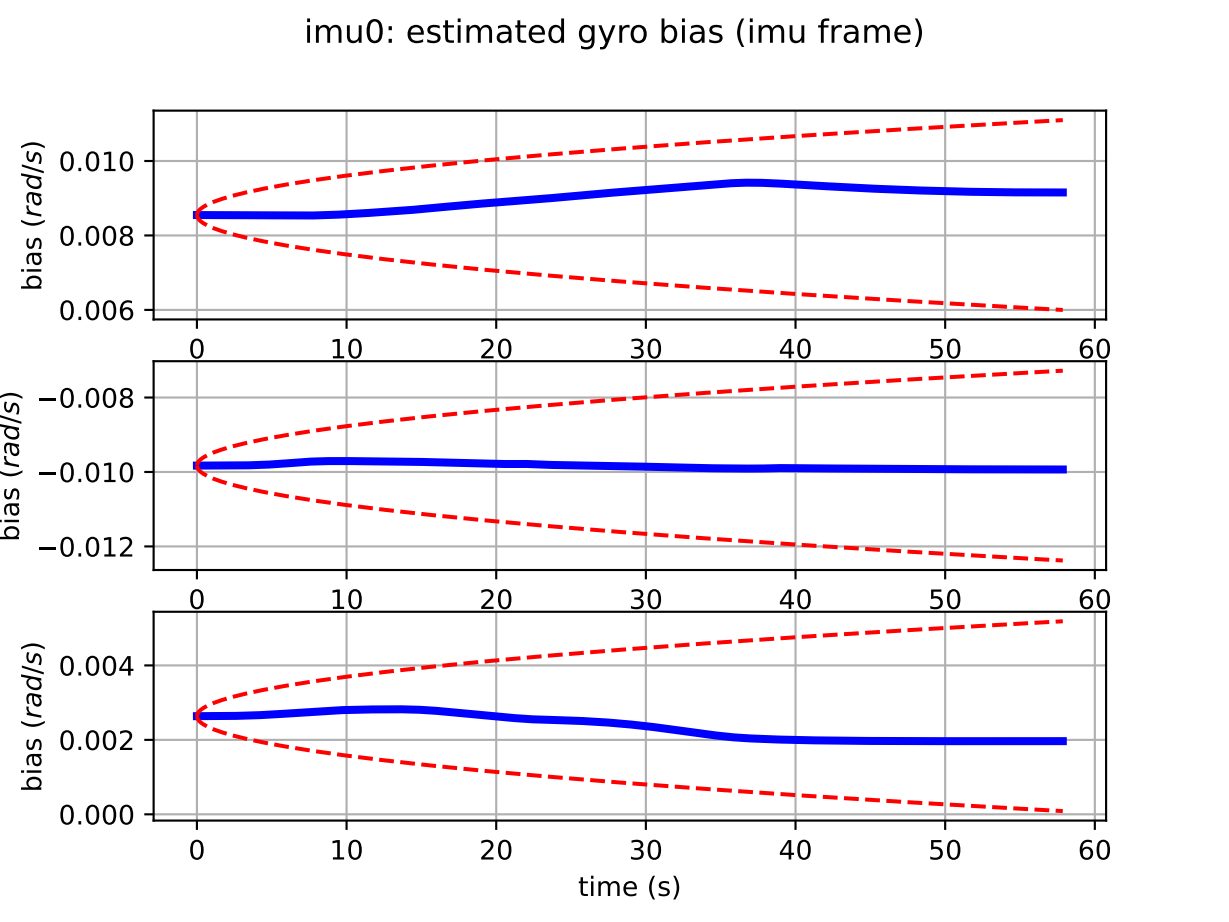

Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
specific force (m/s^2)
specific force (m/s^2)
est. bodyspline
imu0
est. bodyspline
imu0
est. bodyspline
imu0
time (s)

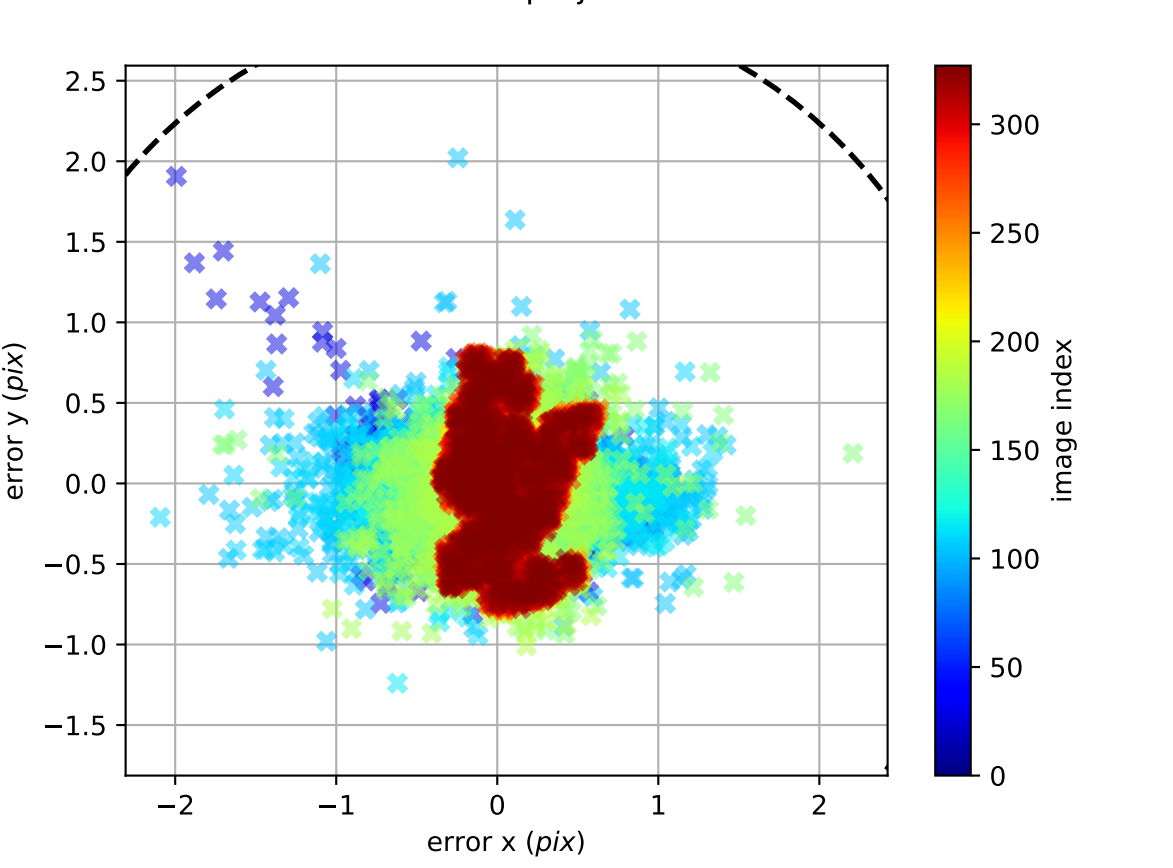

imu0: acceleration error
error (m/s^2)
1
0
−1
0
2000
4000
6000
8000
10000
error (m/s^2)
1
0
−1
0
2000
4000
6000
8000
10000
error (m/s^2)
2.5
0.0
−2.5
0
2000
4000
6000
8000
10000
error index


imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
0.00
−0.25
0
10
20
30
40
50
60
bias (m/s^2)
0.00
−0.25
0
10
20
30
40
50
60
bias (m/s^2)
0.50
0.25
0.00
0
10
20
30
40
50
60
time (s)


Comparison of predicted and measured angular velocities (body frame)
ang. velocity (rad/s)
ang. velocity (rad/s)
ang. velocity (rad/s)
est. bodyspline
imu0
est. bodyspline
imu0
est. bodyspline
imu0
time (s)


imu0: angular velocities error
error (rad/s)
0.1
0.0
−0.1
0
2000
4000
6000
8000
10000
error (rad/s)
0.1
0.0
−0.1
0
2000
4000
6000
8000
10000
error (rad/s)
0.05
0.00
−0.05
0
2000
4000
6000
8000
10000
error index


imu0: estimated gyro bias (imu frame)
bias (rad/s)
0.010
0.008
0.006
0
10
20
30
40
50
60
bias (rad/s)
−0.008
−0.010
−0.012
bias (rad/s)
0.004
0.002
0.000
time (s)


cam0: reprojection errors
error x (pix)
error y (pix)
image index
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2
−1
0
1
2
300
250
200
150
100
50
0